

ESTIMATION ERROR IN PORTFOLIO OPTIMISATION

Bayesian Shrinkage, CVaR and Wasserstein Distributional Robustness

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Abstract

I compare five monthly portfolio rules across twelve value-weighted US industry portfolios: equal weighting, sample mean-variance optimisation, empirical-Bayes/Ledoit-Wolf mean-variance optimisation, historical 95% Conditional Value-at-Risk (CVaR) optimisation, and Wasserstein robust CVaR. The study uses a nested 60-month walk-forward design from January 2015 to September 2025. Hyperparameters are selected before each test month, and each strategy pays 10 basis points per unit of one-way turnover.

Principal findings

- The 1/N benchmark achieved the highest compound return, 11.43% per year, and terminal wealth of 3.20 from an initial dollar.
- Sample MVO was fragile: it returned 7.63%, suffered a -29.6% maximum drawdown, and turned over 22.0% per month.
- Shrinkage MVO cut annualised volatility to 12.3% and improved maximum drawdown to -19.3%, but did not recover the benchmark's return.
- Wasserstein robust CVaR produced the highest zero-rate Sharpe ratio among the optimisation methods (0.766) and a more diversified allocation, but its Sharpe difference from 1/N was not statistically distinguishable from zero.

Interpretation The modern methods mainly improved risk control and stability. They did not establish a statistically reliable return advantage over naive diversification.

1. Research question and motivation

Markowitz (1952) frames investment choice as a trade-off between expected return and covariance risk. In practice, both inputs are estimated with error, and optimisation can turn small errors into large changes in portfolio weights.

The comparison adds empirical-Bayes mean shrinkage, covariance shrinkage, a lower-tail CVaR objective, and Wasserstein robustness to the classical model.

1.1 Testable research question

Research question When the data-generating distribution is estimated from a short rolling history, do shrinkage and distributionally robust tail-risk models generate better out-of-sample portfolios than sample Markowitz optimisation and equal weighting?

1.2 Prior evidence

Ledoit and Wolf (2004) show why a convex combination of the sample covariance matrix and a structured target can be better conditioned than the raw sample estimator. Rockafellar and Uryasev (2000) provide a linear-programming representation of CVaR, making tail-risk optimisation computationally practical. Mohajerin Esfahani and Kuhn (2018) show that data-driven optimisation over Wasserstein ambiguity sets can often be reformulated as tractable convex programmes. DeMiguel, Garlappi and Uppal (2009), however, demonstrate that 1/N is difficult to beat out of sample because estimation error can offset the theoretical gains from optimisation.

1.3 Pre-specified hypotheses

- H1: sample MVO will exhibit the highest concentration and turnover.
- H2: mean and covariance shrinkage will reduce realised volatility and drawdown relative to sample MVO.
- H3: CVaR-based portfolios will improve lower-tail risk relative to variance-based portfolios.
- H4: Wasserstein robustness will reduce concentration relative to historical CVaR.
- H5: no advanced model will show a statistically reliable return advantage over 1/N after costs.

2. Data and experimental protocol

The asset universe is the twelve value-weighted US industry portfolios from the Kenneth R. French Data Library: non-durables, durables, manufacturing, energy, chemicals, business equipment, telecommunications, utilities, shops, health, money and other. The bundled research vintage contains 189 contiguous monthly observations from January 2010 to September 2025. Returns are total percentage returns converted to decimals.

Design element	Specification
Universe	12 US industry portfolios
Data frequency	Monthly
Input sample	2010-01 to 2025-09
Out-of-sample period	2015-01 to 2025-09 (129 months)
Estimation window	Rolling 60 months
Internal validation	Final 12 months of each estimation window
Portfolio constraints	Long-only; sum to 1; 35% cap per industry
Trading cost	10 bps per unit of one-way turnover
Tail probability	95% CVaR

2.1 Nested walk-forward procedure

- At each test month, take only the immediately preceding 60 observations.
- Use the first 48 months to fit candidate models and the final 12 months to select MVO risk aversion or the Wasserstein radius by realised validation Sharpe.
- Refit the selected specification on all 60 observations, fix the portfolio weights, and then reveal the next monthly return.
- Calculate turnover from drifted pre-trade weights, deduct costs, and roll forward by one month.

2.2 Bias controls

The test month is never used for model selection. Parameters are not tuned on the final sample, and all reported returns are net of trading costs. Equal weight uses the same cost convention. The report uses a fixed French data vintage because CRSP revisions can change historical results.

Measurement convention The reported Sharpe statistic assumes a zero monthly risk-free rate and is labelled 'zero-rate Sharpe'. This avoids mixing the portfolio comparison with a separately sourced interest-rate series.

3. Statistical estimators and allocation rules

3.1 Classical sample mean-variance optimisation

For monthly return vector r_t , the classical model estimates the sample mean μ_{hat} and covariance Σ_{hat} and solves a long-only quadratic programme. Gamma is selected in the internal validation period.

$$\min_w \quad 0.5 \gamma w' \Sigma_{\text{hat}} w - \mu_{\text{hat}}' w$$

$$\text{Subject to } 1'w = 1 \text{ and } 0 \leq w_i \leq 0.35.$$

Expected-return errors enter linearly, while covariance inversion and optimisation can amplify noise. The cap prevents a single-industry solution but does not eliminate concentration.

3.2 Empirical-Bayes mean shrinkage

Each industry's sample mean is shrunk toward the cross-sectional grand mean. Let m_i be the sample mean, v_i/T its sampling variance, m_0 the grand mean and τ^2 the estimated between-industry prior variance. The posterior approximation is:

$$\mu_i^B = B_i m_i + (1-B_i)m_0, \quad B_i = \tau^2 / (\tau^2 + v_i/T)$$

Noisy industries receive more shrinkage; more precise estimates retain more of their sample signal.

3.3 Ledoit-Wolf covariance shrinkage

$$\Sigma_{\text{LW}} = \delta F + (1-\delta) \Sigma_{\text{hat}}$$

The target F is a scaled identity in the implementation provided by scikit-learn, and the shrinkage intensity δ follows the Ledoit-Wolf formula. The resulting covariance is positive definite and better conditioned. Shrinkage MVO combines μ^B and Σ_{LW} in the same constrained quadratic programme as sample MVO.

Comparison Sample and shrinkage MVO use the same objective and constraints, isolating the effect of regularised inputs.

4. Tail risk and distributional robustness

4.1 Historical mean-CVaR

For portfolio loss $L_t = -w'r_t$, CVaR is represented using an auxiliary threshold η and non-negative excess-loss variables u_t . The portfolio minimises empirical 95% CVaR subject to an estimated-return floor equal to the empirical-Bayes expected return of the equal-weight portfolio.

$$\min_{\{w, \eta, u\}} \eta + [1/((1-\alpha)T)] \sum_t u_t$$

$$u_t \geq -w'r_t - \eta; \quad u_t \geq 0; \quad (\mu^A)'w \geq (\mu^B)'(1/N)$$

The return floor prevents the optimiser from reducing tail risk by accepting a lower estimated return than the benchmark. SciPy/HiGHS solves the linear programme.

4.2 Wasserstein robust CVaR

Historical CVaR treats the empirical distribution as exact. The robust model instead considers all distributions inside a 1-Wasserstein ball of radius ϵ around it. With an L1 ground metric and linear portfolio loss, the worst-case CVaR adds an infinity-norm concentration penalty:

$$\text{Robust CVaR} = \text{Empirical CVaR} + [\epsilon/(1-\alpha)] \|w\|_{\infty}$$

For long-only weights, $\|w\|_{\infty}$ is linearised by introducing m with $m \geq w_i$. The radius ϵ is chosen on the internal validation block from 0.0005, 0.001, 0.002 and 0.004. Its median selected value was 0.002 monthly-return units.

4.3 Implementation and validation

- Python package with separate data, estimation, optimisation, backtest, metrics and plotting modules.
- Quadratic programmes solved by SLSQP; CVaR programmes solved by HiGHS linear programming.
- Unit tests check data continuity, covariance positive semidefiniteness, full investment, long-only bounds and the robust concentration penalty.
- All 129 monthly fits completed without an optimiser failure.

5. Main out-of-sample results

Strategy	Return	Vol.	Sharpe	Max DD	CVaR loss
Equal Weight	11.4%	15.6%	0.77	-23.2%	9.3%
Sample MVO	7.6%	16.5%	0.53	-29.6%	9.2%
Shrinkage MVO	8.1%	12.3%	0.70	-19.3%	7.7%
Historical Mean-CVaR	9.5%	13.2%	0.75	-21.1%	7.7%
Wasserstein Robust CVaR	9.9%	13.6%	0.77	-23.3%	8.3%

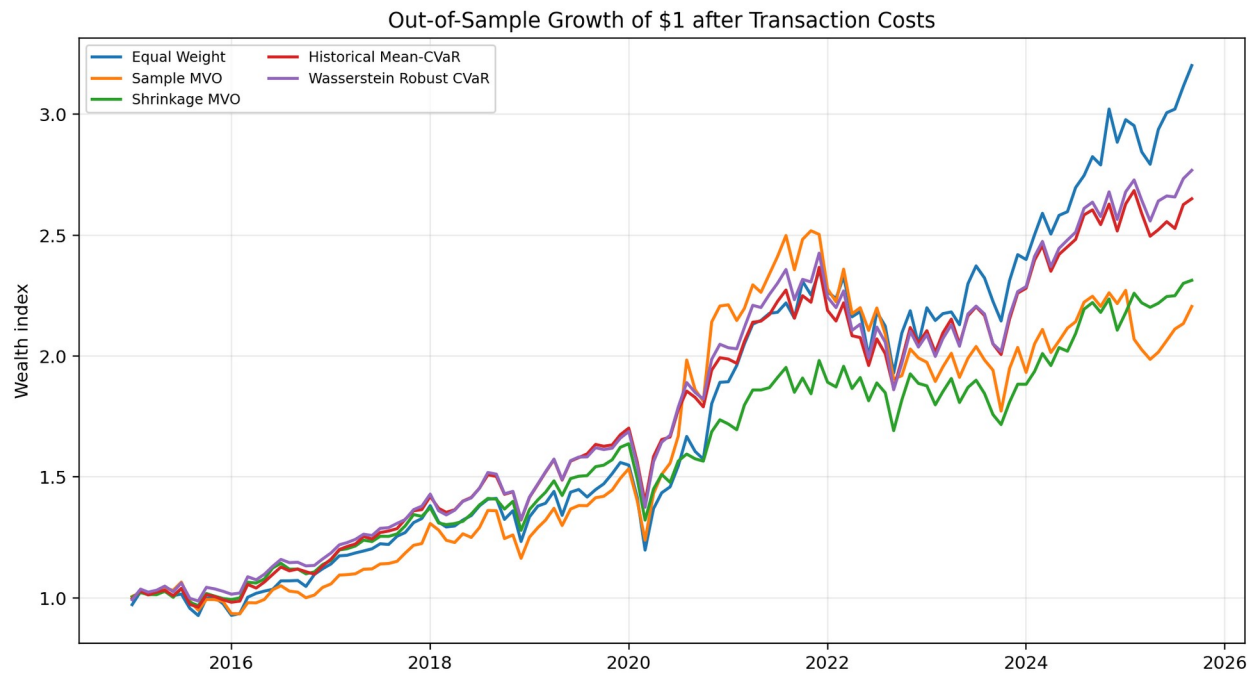


Figure 1. Growth of one dollar after monthly turnover costs, January 2015-September 2025.

Equal weight finished with terminal wealth of 3.20 and the highest compound return. Sample MVO finished at 2.21 with slightly more annualised volatility than 1/N. Shrinkage MVO had the lowest volatility and best maximum drawdown. Both CVaR strategies improved risk control relative to sample MVO while retaining more upside than shrinkage MVO.

Result Controlling estimation error and the risk objective mattered more than maximising the in-sample mean-variance criterion.

6. Downside risk and economic behaviour

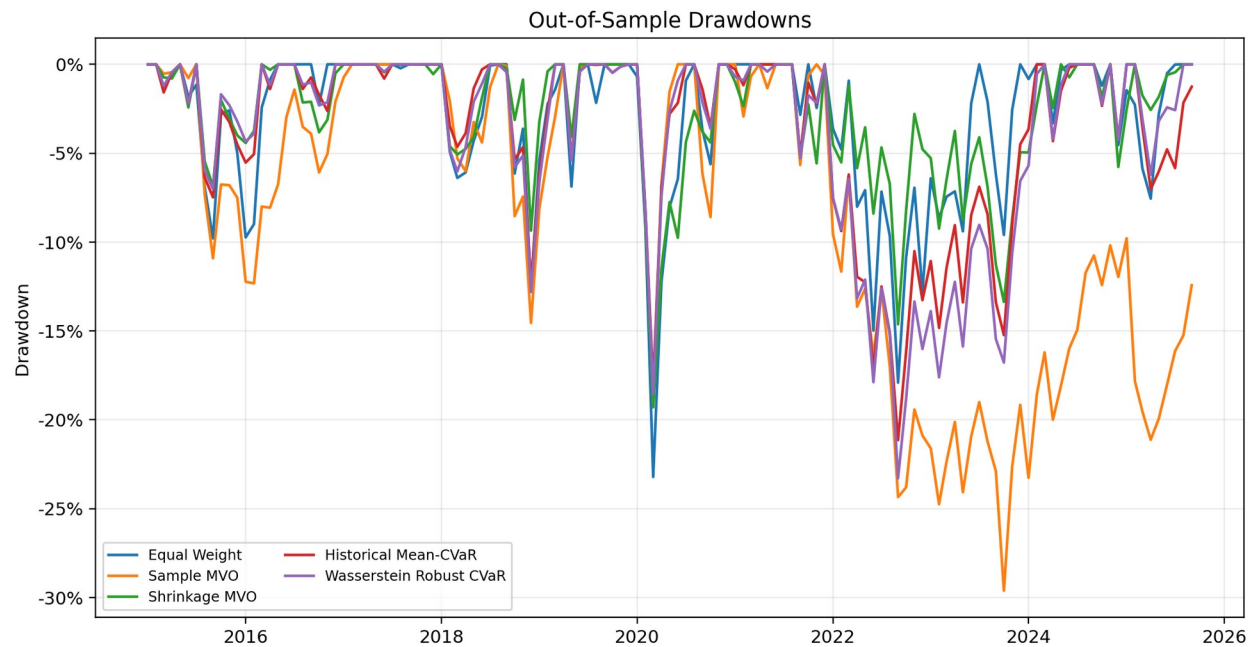


Figure 2. Drawdown paths based on net monthly returns.

Sample MVO experienced the deepest drawdown at -29.6%. Shrinkage MVO limited its maximum drawdown to -19.3%, while historical CVaR reached -21.1%. The estimated 95% monthly CVaR loss fell from 9.3% for equal weight to 7.7% for shrinkage MVO and 7.7% for historical CVaR.

6.1 Stress episodes

Strategy	COVID sell-off	2022 tightening
Equal Weight	-22.7%	-10.9%
Sample MVO	-19.3%	-23.3%
Shrinkage MVO	-19.3%	-8.2%
Historical Mean-CVaR	-18.0%	-16.2%
Wasserstein Robust CVaR	-18.6%	-18.8%

Historical CVaR lost 18.0% in the February-March 2020 sell-off, compared with 22.7% for 1/N. During the 2022 tightening episode, shrinkage MVO performed best (-8.2%) and robust CVaR lost 18.8%. Distributional robustness does not protect against every regime.

7. Turnover, concentration and regime dependence

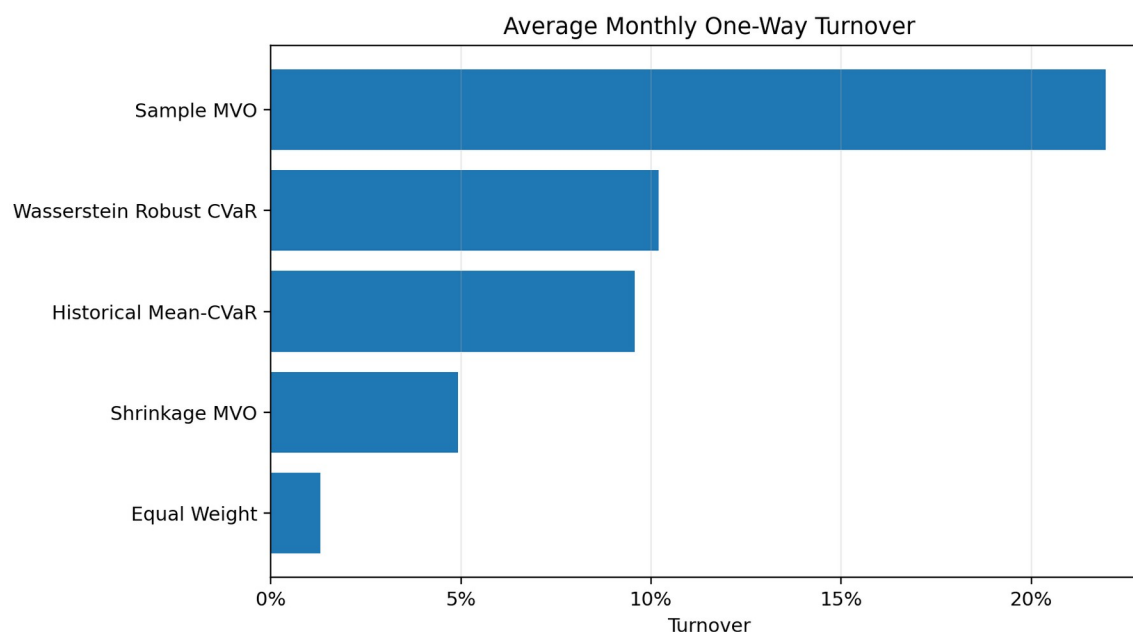


Figure 3. Average monthly one-way turnover. Ten basis points are charged per unit of turnover.

Strategy	Turnover	Cost drag	Effective N	Avg. max wt.
Equal Weight	1.3%	0.02%	12.0	8.3%
Sample MVO	22.0%	0.26%	3.5	34.4%
Shrinkage MVO	4.9%	0.06%	4.0	34.4%
Historical Mean-CVaR	9.6%	0.11%	3.6	35.0%
Wasserstein Robust CVaR	10.2%	0.12%	4.8	24.4%

Sample MVO reallocated 22.0% per month on average and held an effective 3.5 industries. Its average largest position was close to the 35% cap. Wasserstein robustness increased effective breadth to 4.8 industries and reduced the average largest position to 24.4% because the L1 ambiguity penalty directly penalises the largest weight.

7.1 Volatility regimes

Strategy	High-vol. Sharpe	Lower-vol. Sharpe
Equal Weight	0.90	0.64
Sample MVO	0.74	0.24
Shrinkage MVO	0.67	0.78
Historical Mean-CVaR	0.91	0.57
Wasserstein Robust CVaR	0.89	0.63

Historical CVaR had the strongest high-volatility Sharpe (0.91), narrowly above equal weight (0.90). Shrinkage MVO performed better in lower-volatility months (0.78) than in high-volatility months (0.67). Sample MVO's lower-volatility Sharpe was only 0.24, consistent with unstable mean estimates rather than compensation for systematic stress exposure.

8. Statistical inference, robustness and limitations

A 2,000-draw circular moving-block bootstrap with six-month blocks preserves short-run dependence. Each optimisation strategy is compared with equal weight using annualised arithmetic mean-return and zero-rate Sharpe differences. Confidence intervals are percentile intervals.

Strategy vs 1/N	Mean diff.	95% CI	Sharpe diff.	95% CI
Sample MVO	-3.4%	[-8.2%, 1.7%]	-0.25	[-0.58, 0.04]
Shrinkage MVO	-3.5%	[-7.3%, 0.1%]	-0.08	[-0.33, 0.17]
Historical Mean-CVaR	-2.1%	[-5.5%, 0.8%]	-0.02	[-0.26, 0.17]
Wasserstein Robust CVaR	-1.7%	[-4.7%, 1.1%]	-0.01	[-0.21, 0.18]

Every mean-return difference was negative. The confidence intervals for historical and robust CVaR included zero, while the shrinkage MVO interval ended close to zero. No Sharpe-difference interval showed a positive advantage. Robust CVaR's observed Sharpe was 0.008 below 1/N, with a 47.3% bootstrap probability of a positive difference. This supports non-superiority, not equality.

8.1 Limitations

- Industry portfolios are research constructs rather than directly investable securities; implementation costs for actual products may differ.
- The universe excludes bonds, cash and international assets, so results are not a complete strategic asset-allocation study.
- Sixty monthly observations leave only a few empirical observations in the 5% CVaR tail, increasing sampling uncertainty.
- The Wasserstein radius is validation-selected rather than calibrated as a formal confidence radius.
- A zero risk-free rate is used for the labelled Sharpe statistic, and inference is sensitive to block length and sample length.
- Historical backtests cannot establish future profitability; model and data vintages may change.

Reproducibility The repository stores monthly weights, returns, turnover, selected hyperparameters, bootstrap results and figures.

9. Conclusion

Sample MVO performed poorly because noisy means and covariances produced concentrated, high-turnover decisions. Empirical-Bayes and Ledoit-Wolf shrinkage improved realised risk. CVaR shifted the objective from variance to tail losses and performed better in the COVID sell-off.

Wasserstein robustness produced a broader portfolio than historical CVaR and the strongest Sharpe among the optimisation methods, but it did not beat equal weighting on compound return or statistical inference. The robust penalty therefore helped control concentration and distributional uncertainty without solving the more fundamental problem of forecasting expected returns. Equal weighting remained the strongest overall benchmark.

Financial models should separate estimation, decision and evaluation. A sophisticated objective has value only if its decisions hold up on unseen data, after costs and under stress.

References

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Repository deliverables

The accompanying repository includes the versioned data snapshot, official-data downloader, tested Python package, walk-forward runner, all output CSV files, figures, editable Word report and PDF report.